

ARUNENDRA TRIPATHI

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EDUCATION

Indian Institute of Information Technology Manipur

2023 – Present

Bachelor of Technology (Computer Science and Engineering)

Manipur, India

Relevant Coursework: Data Structures & Algorithms, Machine Learning, Computer Vision, Operating Systems, DBMS, Computer Networks

INTERNSHIPS

1) Dciphers IT Solutions (P) Ltd.

Greater Noida, India

Software Development Intern

May 2026 – Jul 2026

- Built robust backend services in Go (Golang) for the core monitoring engine, enabling automated discovery of devices across Fortinet firewalls and Wi-Fi networks with real-time monitoring of key device metrics.
- Designed responsive UIs for the Database Access Management System and developed the Forms Management module for the Dciphers Corporate Website.

Internship Project: Network Management System (NMS) | Go, Fiber, PostgreSQL, TimescaleDB, React.js, TypeScript, SNMP, JWT, Docker

- Developed a full-stack NMS with a high-performance Go + Fiber backend implementing REST APIs, JWT authentication, and role/permission-based access control.
- Integrated SNMP for automated device discovery and metric collection, PostgreSQL and TimescaleDB for configuration and time-series data, and a React + TypeScript dashboard for topology visualization, device inventory, and CSV reporting.

2) Indian Institute of Information Technology (IIIT) Vadodara

Vadodara, India

Research Intern – Reversible Data Hiding & Image Security

Jan 2026 – Present

- Developed CNN-based prediction models for Reversible Data Hiding (RDH) using Cross and Dot partitioning techniques, and evaluated CNRP against DP, MEDP, GAP, and BIP through MSE and PSNR-based benchmarking.
- Implemented Histogram Shifting and Prediction Error Expansion for reversible data embedding.

Research Project: Reversible Data Hiding-Based Contrast Enhancement for Medical Images

- Worked on reversible data hiding algorithm for medical image enhancement, integrating superpixel segmentation, multi group histogram stretching, and encrypted image steganography techniques.
- Applied deep learning-based ROI segmentation and implemented evaluation pipelines to measure embedding capacity, image quality, and recovery accuracy.

3) National Institute of Technology (NIT) Warangal

Warangal, India

Research Intern – Machine Learning

Dec 2025 – Jan 2026

- Designed and evaluated machine learning driven optimization techniques for resource allocation and workload management in fog and cloud computing environments.
- Implemented and analyzed Crow Search Algorithm (CSA) variants to improve optimization efficiency across distributed computing scenarios.

Research Project: ScholarSynth (Research Paper Analysis Platform)

[Link](#)

- Built an AI-powered research paper analysis platform using React, TypeScript, Vite, Tailwind CSS, KaTeX, and Google Gemini 2.5 Flash to automate extraction of key research insights from PDF documents.

PROJECTS

1) CodeOrbit : Unified Coding Performance Dashboard

[Link](#)

- Full stack coding analytics platform using Next.js, React, TypeScript, Node.js, Express.js, MongoDB, and Firebase Authentication to unify coding performance data in one dashboard.
- Integrated coding statistics from LeetCode, Codeforces, GitHub, CodeChef, and GeeksforGeeks, enabling centralized performance tracking across multiple platforms.
- Integrated LLaMA 3.3 70B (Groq API) to provide personalized coding recommendations, while developing responsive Tailwind CSS interfaces and optimizing API performance.
- Implemented automated synchronization workflows, 365-day activity heatmaps, consistency scoring, and long-term progress analytics.

TECHNICAL SKILLS

AI/ML Technologies: Machine Learning, Deep Learning, Computer Vision, CNN-Based Predictive Modeling, Optimization Algorithms (Crow Search Algorithm), Reversible Data Hiding, Image Security

LLM & AI Systems: Prompt Engineering, Semantic Search, AI Content Generation, API Integration, AI Chat Workflows

Libraries & Frameworks: NumPy, Pandas, OpenCV, Scikit-learn, PyTorch, TensorFlow, Vite, KaTeX

Programming Languages: Python, C, C++, Go (Golang), JavaScript, TypeScript

Frontend Development: React.js, Next.js, TypeScript, Tailwind CSS, Responsive UI Development

Backend Development: Node.js, Express.js, REST API Development, Authentication Systems, Server Side Development

Databases: PostgreSQL, MongoDB, SQL, Database Design and Management

Version Control & Collaboration: Git, GitHub, Open Source Contribution Workflows

Core Computer Science: Data Structures and Algorithms, OOP, Performance Analysis and Optimization

ACHIEVEMENTS

- McKinsey & Company Virtual Experience Program (2025):** Solved real-world consulting case problems using structured, data-driven approaches.
- GirlScript Summer of Code (GSSoC):** Selected Open-Source Contributor.
- LeetCode:** Solved 150++ questions and earned the 100 Days Badge for consistent problem-solving.
- Hacktoberfest 2024:** Earned contribution badges with multiple merged pull requests.
- Open-Source Merged PRs:** Redis Python client (redis-py) and Red Hat Podman container-lib.
- Kaggle:** Earned the Competitor badge and multiple activity streak badges.